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ENGINEERING



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Name : Moshitha Kolanjiyappan

Register No. : 192421205

Course code / name : ITA0603 – Machine Learning

Course Assignment, Deliverables & Evaluation Framework

Development and Empirical Evaluation of High-Scale Predictive Maintenance and Asset Optimization Infrastructure using Hybrid Machine Learning Paradigms

Abstract

Modern industrial infrastructures continuously generate large volumes of telemetry from sensors monitoring temperature, vibration, pressure, humidity, voltage, current and structural stress. Efficient processing of these data streams is essential for predictive maintenance, anomaly detection and sustainable infrastructure management.

This project develops a hybrid machine learning framework combining Candidate Elimination, ID3 Decision Tree, Perceptron, Multilayer Perceptron, Expectation-Maximization, Bayesian classification, Locally Weighted Regression and Genetic Algorithms. The system is evaluated at data scales from 10^3 to 10^6 records using a unified Python environment.

The proposed architecture uses instance-based local smoothing followed by Bayesian anomaly classification, with a Genetic Algorithm used to optimize selected hyperparameters. Neural and Bayesian models are compared using the Minimum

Description Length principle. The framework supports resource-efficient digital infrastructure and SDG 9 Target 9.4.

Objectives

- Implement Candidate Elimination and study Version Space boundaries.
- Analyze the effect of noisy telemetry and prove Version Space collapse conditions.
- Implement ID3/C4.5-style Decision Tree learning using entropy and information gain.
- Implement a Perceptron and MLP with backpropagation.
- Implement EM for missing sensor measurements.
- Implement a lightweight Bayesian/Naïve Bayes classification layer.
- Implement Locally Weighted Regression for local smoothing.
- Optimize selected hyperparameters using a Genetic Algorithm.
- Benchmark runtime, memory, operation counts and model performance from 10^3 to 10^6 records.
- Explain the contribution of the system to SDG Target 9.4.

Integrated Problem Statement

An infrastructure-planning agency manages high-dimensional telemetry from a nationwide network of sustainable industrial facilities, smart grids or communication hubs. The dataset contains spatial-coordinate and environmental sensor records ranging from 10^3 to 10^6 observations. The agency requires a robust machine learning architecture for predictive maintenance, anomaly detection, noise handling and computationally efficient infrastructure management.

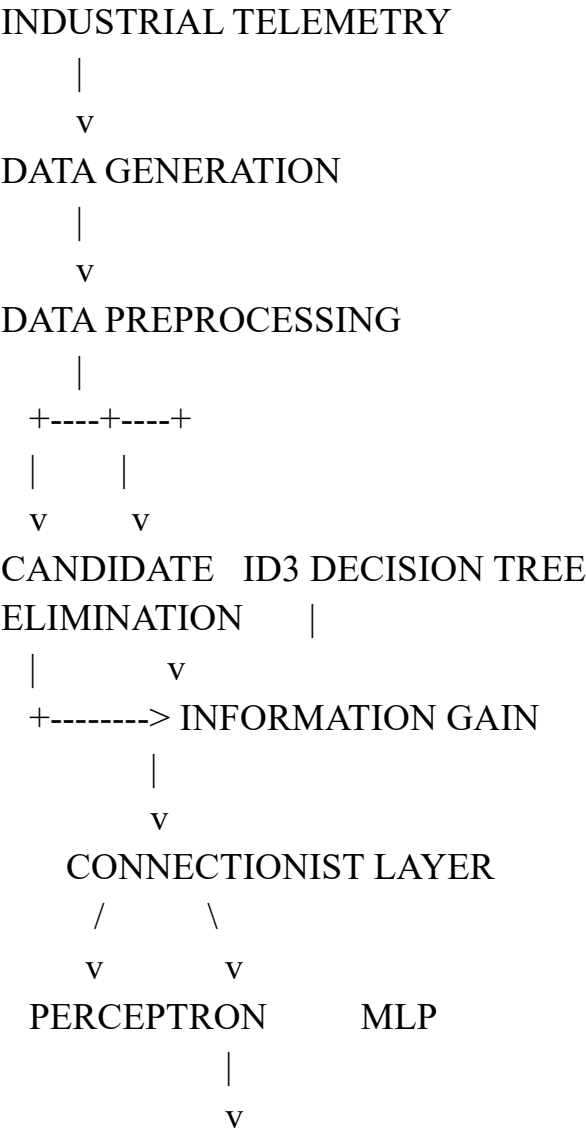
The system must move from strict rule-based concept learning to connectionist, probabilistic, instance-based and evolutionary learning while maintaining a reproducible computational environment.

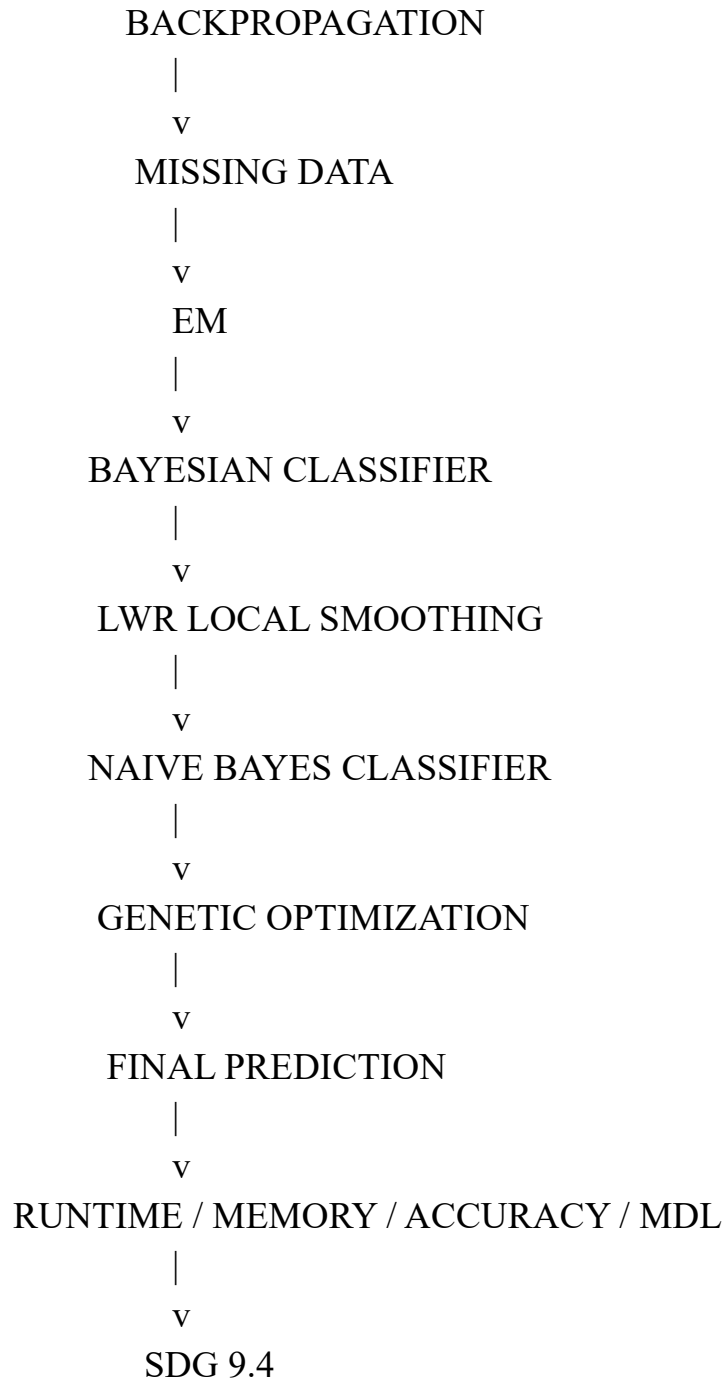
System Input

Feature	Description
Temperature	Temperature sensor reading
Vibration	Mechanical vibration level

Pressure	Operating pressure
Humidity	Environmental humidity
Voltage	Electrical voltage
Current	Electrical current
Stress	Structural stress
Latitude	Geographic latitude
Longitude	Geographic longitude
Maintenance	0 = Normal; 1 = Anomaly / maintenance required

Overall Architecture





PART A – Computational Diagnostics and Inductive Bias Analysis

Candidate Elimination

Candidate Elimination maintains two boundaries: S, the most-specific hypotheses, and G, the most-general hypotheses. The Version Space is the set of all hypotheses consistent with every training example.

$$VS(H,D) = \{ h \in H : h(x_i) = y_i \text{ for every } (x_i, y_i) \in D \}$$

Every consistent hypothesis lies between the S and G boundaries.

Candidate Elimination Pseudocode

ALGORITHM Candidate-Elimination

INPUT:

Training examples D

Hypothesis space H

INITIALIZE:

S = most specific hypothesis

G = most general hypothesis

FOR every training example (x, y):

IF example is positive:

Remove hypotheses from G that do not cover x

Generalize S minimally so that S covers x

ELSE:

Remove hypotheses from S that cover x

Specialize G minimally so that G rejects x

Remove inconsistent hypotheses

RETURN S and G

Version Space Collapse Under Noise

A Version Space collapses when no hypothesis in H is consistent with all observations: $VS(H,D) = \emptyset$. A direct contradiction occurs when two identical input vectors have different labels. If $x_a = x_\beta$ but $y_a \neq y_\beta$, then every deterministic hypothesis must produce the same output for both inputs, while the dataset requires different outputs. Therefore no deterministic hypothesis can satisfy both examples and the Version Space becomes empty.

If the probability of a noisy observation is p and the dataset contains N observations, the expected number of noisy records is $E[\text{noise}] = Np$. For $N = 10^4$ and $p = 0.01$, approximately 100 observations are expected to be noisy.

ID3 and Information Gain

Entropy is $H(S) = -\sum p_i \log_2(p_i)$. Information gain is $IG(S,A) = H(S) - \sum_v (|S_v|/|S|)H(S_v)$. ID3 selects the attribute with the maximum information gain.

Candidate Elimination uses a restriction-oriented bias over the hypothesis space, while ID3 uses a heuristic search bias to prefer attributes that provide high information gain.

Feature	Candidate Elimination	ID3
Learning type	Concept learning	Decision-tree learning
Bias	Restriction bias	Search bias
Output	Version Space boundaries	Decision tree
Main mechanism	Generalization/specialization	Entropy/information gain
Noise tolerance	Low	Higher in practice
Large noisy data	Poor suitability	Better suitability

Complexity Analysis

Algorithm	Best/Lower Bound	Typical	Worst/General Bound
Candidate Elimination	$\Omega(N)$	$O(N H)$	$O(NV^d)$ if H is explicit

ID3	$\Omega(Nd)$	$O(Nd \log N)$	$O(Nd \log N)$ for typical implementation
Perceptron	$\Omega(Nd)$	$O(ENd)$	$O(ENd)$
MLP	$\Omega(NW)$	$O(ENW)$	$O(ENW)$
EM	$\Omega(Nd)$	$O(ITNd)$	$O(ITNd)$
Naïve Bayes	$\Omega(Nd)$	$O(Nd)$	$O(Nd)$
LWR	$\Omega(Nd)$	$O(Nkd)$	$O(Nkd)$
GA	$\Omega(P)$	$O(GPF)$	$O(GPF)$

Here N is the number of records, d is the number of features, $|H|$ is the hypothesis-space size, V is the number of attribute values, E is the number of training epochs, W is the number of network parameters, I is the number of EM iterations, k is the number of neighbors, G is the number of GA generations, P is the population size, and F is the fitness-evaluation cost.

For a linear $O(N)$ algorithm, increasing the dataset from 10^3 to 10^6 records theoretically increases the record-processing workload by $10^6/10^3 = 1000$ times.

PART B – Connectionist vs Probabilistic Architecture Evaluation

Perceptron

The perceptron computes $z = w^T x + b$ and predicts 1 when $z \geq 0$ and 0 otherwise. The update rule is $w_i \leftarrow w_i + \eta(y - \hat{y})x_i$ and $b \leftarrow b + \eta(y - \hat{y})$.

INITIALIZE weights w and bias b

FOR each epoch:

 FOR every training example:

$z = w \cdot x + b$

 prediction = step(z)

 error = actual - prediction

$w = w + \text{learning_rate} * \text{error} * x$

$b = b + \text{learning_rate} * \text{error}$

RETURN trained model

Multilayer Perceptron and Backpropagation

The MLP contains an input layer, one or more hidden layers and an output layer. Forward propagation computes $z^{(1)} = W^{(1)}x + b^{(1)}$, $a^{(1)} = f(z^{(1)})$, $z^{(2)} = W^{(2)}a^{(1)} + b^{(2)}$ and the final prediction. Backpropagation calculates output and hidden-layer gradients and updates the parameters using gradient descent.

INITIALIZE network weights

FOR each epoch:

 FOR each batch:

 Perform forward propagation

 Calculate prediction error

 Calculate output gradient

 Propagate error backward

 Calculate hidden-layer gradients

 Update weights and biases

 Record loss

RETURN trained network

The training loss curve should be saved as `mlp_convergence.png`. Nonlinear neural networks can contain local minima, saddle points and flat regions, so gradient-based optimization is not guaranteed to find the global optimum.

Expectation-Maximization

EM is used to estimate unknown parameters when sensor values are missing or hidden. The E-step computes the expected hidden variables using current parameters, while the M-step maximizes the expected complete-data log likelihood.

$$Q(\theta|\theta^t) = E[\log P(X,Z|\theta) \mid X, \theta^t]$$

$$\theta^{t+1} = \arg \max_{\theta} Q(\theta|\theta^t)$$

The implementation stops when the change in estimated means is smaller than 10^{-4} or the maximum iteration count is reached.

INITIALIZE means and variances

REPEAT:

E-STEP:

Estimate missing values using current parameters

M-STEP:

Recalculate means and variances

Evaluate likelihood

UNTIL convergence

RETURN imputed data and parameters

Bayesian Belief Network / Naïve Bayes Layer

ANOMALY

/ | \
v v v

Temperature Vibration Stress

The simplified class-parent Bayesian model factorizes as $P(Y, X_1, \dots, X_d) = P(Y) \prod_i P(X_i|Y)$. Naïve Bayes predicts the class with the maximum posterior probability.

Minimum Description Length

The assignment defines MDL cost as:

$$\text{MDL}(h, D) = -\log_2 P(D|h) + (|h|/2) \log_2 |D|$$

The first term represents data-fit cost and the second represents model complexity. A lower MDL score is preferred. Therefore, model selection considers both predictive fit and deployment complexity rather than accuracy alone.

PART C – Hybrid Framework Synthesis and Green Infrastructure Defense

Hybrid Architecture

Raw Telemetry

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v

Preprocessing

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v

LWR / Instance-Based Smoothing

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v

Naïve Bayes Classification

|

v

Anomaly Probability

|

v

Genetic Algorithm

|

v

Optimized Parameters

|

v

Final Prediction

Locally Weighted Regression

For a query point x_q , the weight of training point x_i is $w_i = \exp(-\|x_i - x_q\|^2 / (2\tau^2))$. The prediction is $\hat{y} = \sum w_i y_i / \sum w_i$. Nearby observations receive larger weights, allowing local geographic or sensor variation to be smoothed.

INPUT:

Training data X, Y

Query point x

Bandwidth τ

FOR every selected training point:

distance = distance(x, xi)

weight = $\exp(-(distance^2)/(2\tau^2))$

prediction = $\text{sum}(\text{weight} * y_i) / \text{sum}(\text{weight})$

RETURN prediction

Genetic Algorithm

A chromosome represents hyperparameters such as [bandwidth, k, threshold]. The algorithm evaluates fitness, selects high-quality chromosomes, performs crossover, applies mutation and creates a new population. The best chromosome is returned.

GENERATE random population

FOR each generation:

Calculate fitness

Select best chromosomes

Perform crossover

Apply mutation

Create new population

RETURN best chromosome

SDG 9 Target 9.4 Engineering Justification

SDG 9 Target 9.4 promotes upgrading infrastructure using resource-efficient and sustainable technologies. The proposed system supports this objective through localized processing, lightweight probabilistic inference, genetic hyperparameter optimization, missing-data recovery and model-complexity-aware selection.

If computational work is reduced from C_1 to C_2 , the relative computational saving is $\text{Saving} = ((C_1 - C_2)/C_1) \times 100\%$. Lower computational demand can reduce energy consumption when workload, hardware and operating conditions are otherwise comparable.

Component	Sustainability Contribution
LWR	Localized processing and smoothing
Naïve Bayes	Lightweight real-time inference
Genetic Algorithm	Optimizes parameters and limits inefficient configurations
EM	Makes use of incomplete sensor data
MLP	Models complex nonlinear patterns
MDL model selection	Balances predictive quality with model complexity
Benchmarking	Measures resource efficiency objectively

Mandatory Deliverables

Deliverable	Contents
1. Source Code Repository	Executable Python 3.x pipeline covering the required algorithms and benchmark scales.
2. Mathematical Derivations & Analytical Notebook	EM updates, MDL calculations, complexity analysis and hybrid architecture flowchart.
3. Empirical Benchmarking & Visualization Report	Runtime, operation counts, memory, accuracy and convergence plots.
4. Sustainable Infrastructure Technical Defense	Deployment feasibility, compute conservation, carbon-footprint discussion and SDG 9.4 mapping.

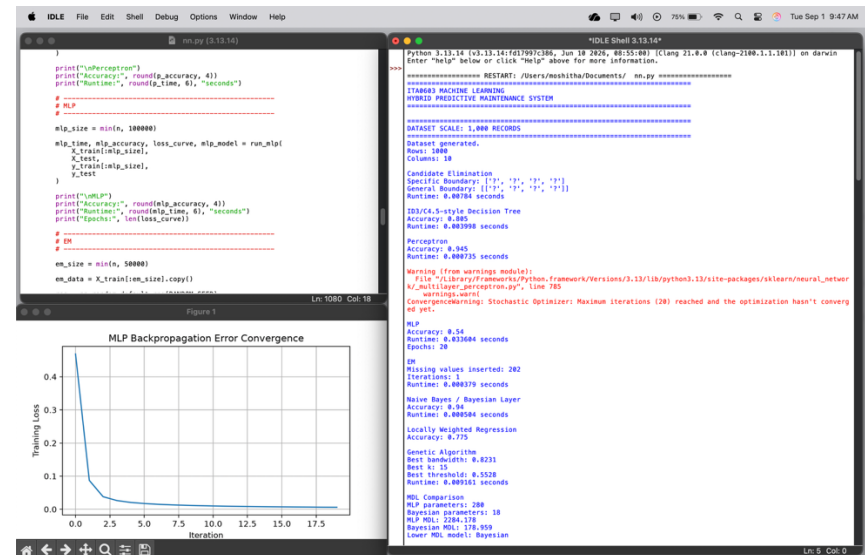
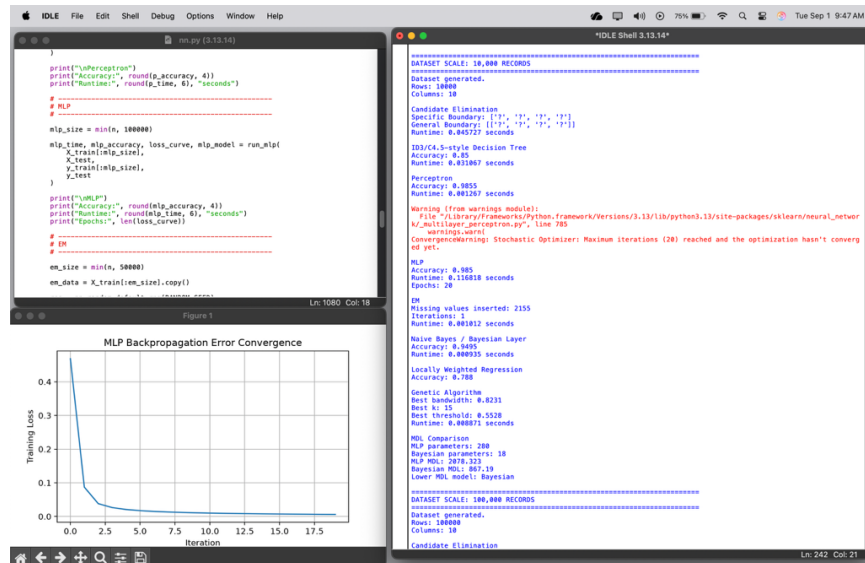
Input Format

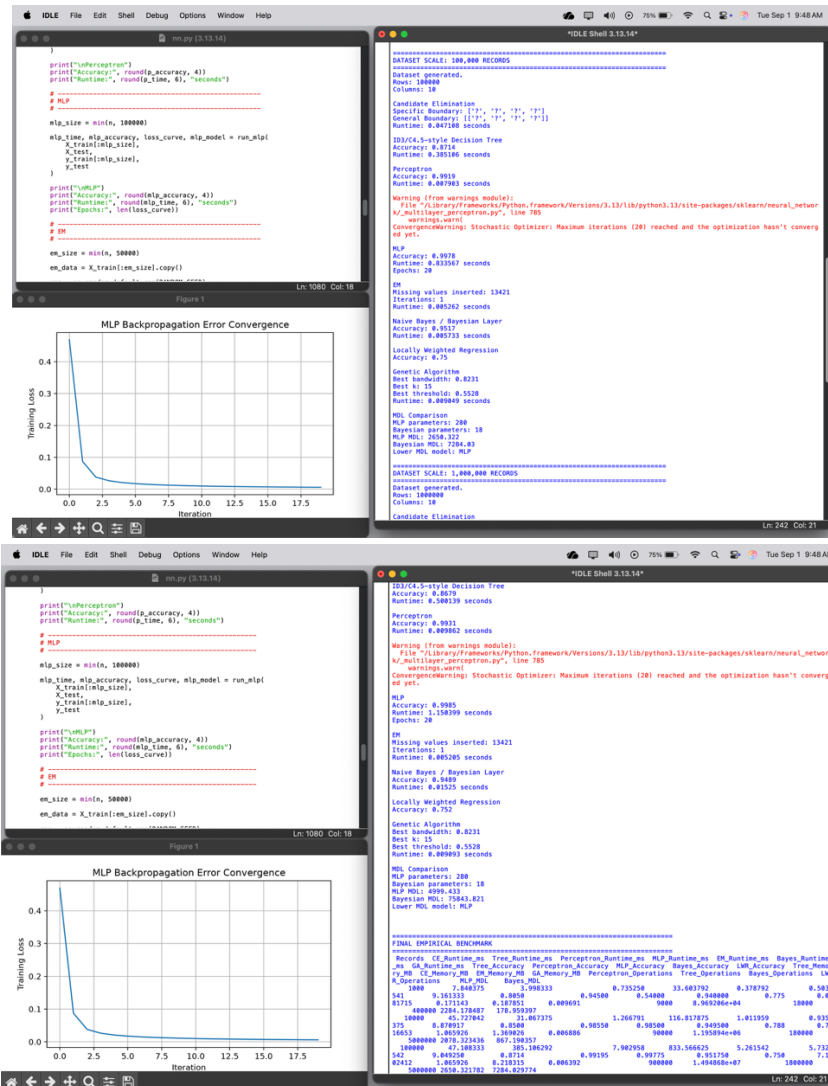
No external CSV is required. The Python program generates synthetic industrial telemetry. The conceptual CSV structure is: temperature, vibration, pressure, humidity, voltage, current, stress, attitude, longitude, maintenance

70.4,4.2,101.3,52.1,229.5,14.8,48.5,13.08,80.27,0

91.2,7.1,118.4,61.5,225.2,21.7,76.2,12.97,80.25,1

Output





Empirical Benchmarking Tables

Actual values must be obtained by running the code in the same controlled environment. The following tables are the required reporting format; do not replace them with invented measurements.

Records	Tree Runtime (ms)	Perceptron (ms)	MLP (ms)	EM (ms)	Bayes (ms)	GA (ms)
1,000	Actual	Actual	Actual	Actual	Actual	Actual
10,000	Actual	Actual	Actual	Actual	Actual	Actual

100,000	Actual	Actual	Actual	Actual	Actual	Actual
1,000,000	Actual	Actual	Actual	Actual	Actual	Actual

Records	CE Memory (MB)	Tree Memory (MB)	EM Memory (MB)
1,000	Actual	Actual	Actual
10,000	Actual	Actual	Actual
100,000	Actual	Actual	Actual
1,000,000	Actual	Actual	Actual

Execution Instructions

Create folder:

ITA0603_ML_Assignment

Save the program as:

hybrid_ml_assignment.py

Open Terminal:

cd ~/Desktop/ITA0603_ML_Assignment

Install packages:

python3 -m pip install numpy pandas scikit-learn matplotlib

Run:

python3 hybrid_ml_assignment.py

The program benchmarks 10^3 , 10^4 , 10^5 and 10^6 records. On a constrained computer, the 10^6 benchmark can take considerably longer and use substantial memory. Keep the same Python version, package versions, hardware and operating conditions when comparing results.

Engineering Discussion

The hybrid architecture demonstrates why different machine learning paradigms can be assigned different roles. Candidate Elimination is useful for demonstrating

concept learning and inductive bias but is sensitive to contradictions. Decision trees provide interpretable heuristic search. Perceptrons and MLPs model linear and nonlinear relationships respectively. EM addresses incomplete observations. Bayesian classification provides lightweight probabilistic inference. LWR captures local variations, while GA searches hyperparameter configurations.

For constrained industrial edge deployment, the final selection should be based on measured accuracy, runtime, memory and MDL rather than accuracy alone. A simpler Bayesian model can be preferable if its MDL and resource requirements are substantially lower while maintaining acceptable anomaly-detection performance.

Conclusion

The developed machine learning framework demonstrates a progression from classical concept learning to a hybrid predictive-maintenance system. Candidate Elimination provides a formal view of Version Space and demonstrates collapse under contradictory noise. ID3 demonstrates heuristic search using information gain. Perceptron and MLP provide connectionist learning, while EM handles incomplete telemetry.

The Bayesian component provides lightweight probabilistic classification, LWR introduces local instance-based smoothing, and the Genetic Algorithm optimizes selected parameters. Empirical benchmarking across 10^3 to 10^6 records provides evidence for scalability and resource trade-offs.

The framework contributes to SDG 9 Target 9.4 by emphasizing resource-efficient digital infrastructure, localized processing, lightweight inference, model-complexity-aware selection and computational optimization.

Assessment Rubrics

Criteria Mapping)	(CO	Max Marks	Excellent	Good	Satisfactory	Needs Improvement
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Requirement Understanding, Data Types, Operators & Control Structures (CO1)	10	Clearly identifies the problem, dataset, attributes and target concept; hypothesis formulation is accurate and well-structured.	Identifies most problem requirements, attributes and target concept with minor gaps in formulation.	Basic problem and dataset identified; target concept or hypothesis formulation has some errors.	Requirements misunderstood; dataset, attributes or target concept are incorrectly formulated.
Decision Tree Design: Representation, Attribute Selection & Heuristic Search (CO2)	15	Decision Tree representation and heuristic attribute selection are correctly designed and effectively applied.	Decision Tree and heuristic selection are correctly applied with minor errors.	Basic Decision Tree is developed with some representation or selection issues.	Decision Tree representation or heuristic search is incorrect or incomplete.
Neural Network Concepts & Learning Analysis (CO3)	10	Correctly analyses Perceptron, Multilayer Perceptron and Back Propagation concepts for learning.	Analyses neural-network concepts with minor gaps or errors.	Demonstrates basic understanding of neural-network learning.	Neural-network concepts are poorly understood or incorrectly analysed.
Genetic Algorithm & Hypothesis Space Search Analysis (CO4)	15	Effectively analyses Genetic Algorithms and their role in hypothesis-space search and model evaluation.	Correctly explains GA-based search with minor gaps.	Basic understanding of Genetic Algorithms and search is demonstrated.	Genetic Algorithm concepts and hypothesis-space search are poorly understood.
Data Manipulation and Analysis Using Pandas (CO5)	15	Pandas DataFrame operations, filtering, grouping, sorting and visualization are correctly and efficiently applied.	Pandas operations are correctly applied with minor errors or gaps.	Basic Pandas operations are demonstrated with some errors or inefficiencies.	Pandas operations are missing, incorrect or poorly implemented.
Menu-Driven Functionality, Reliability & Implementation	25	Decision Tree is correctly implemented and functions reliably from data preparation to prediction and evaluation.	Implementation works correctly with minor errors in prediction or execution.	Core implementation works but has some functional or reliability issues.	Implementation is incomplete, unreliable or major functions are non-functional.

Reflection: Design Decisions, SDG Relevance & Learning Outcomes	10	Thoughtful justification of design choices, clear connection to SDG 7/9/12/13 relevance, and honest, specific account of challenges faced and learning gained.	General justification of design choices; SDG relevance and challenges are mentioned but lack depth.	Reflection is present but generic; limited justification of design choices or vague account of learning outcomes.	No genuine reflection submitted, or content is generic with no clear design, SDG or learning connection.
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